

ME5412 Robotics for Healthcare

Part 2

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Course Instructor



- Dr. Yu Haoyong
 - Associate Professor
- Research areas:
 - Rehab robotics
 - Assistive devices
 - Human robot interaction
 - Bio-inspired robotics
 - System dynamics and intelligent control
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Biorobotics Lab Overview



Construction, logistics,
factories, and farming,
caregivers



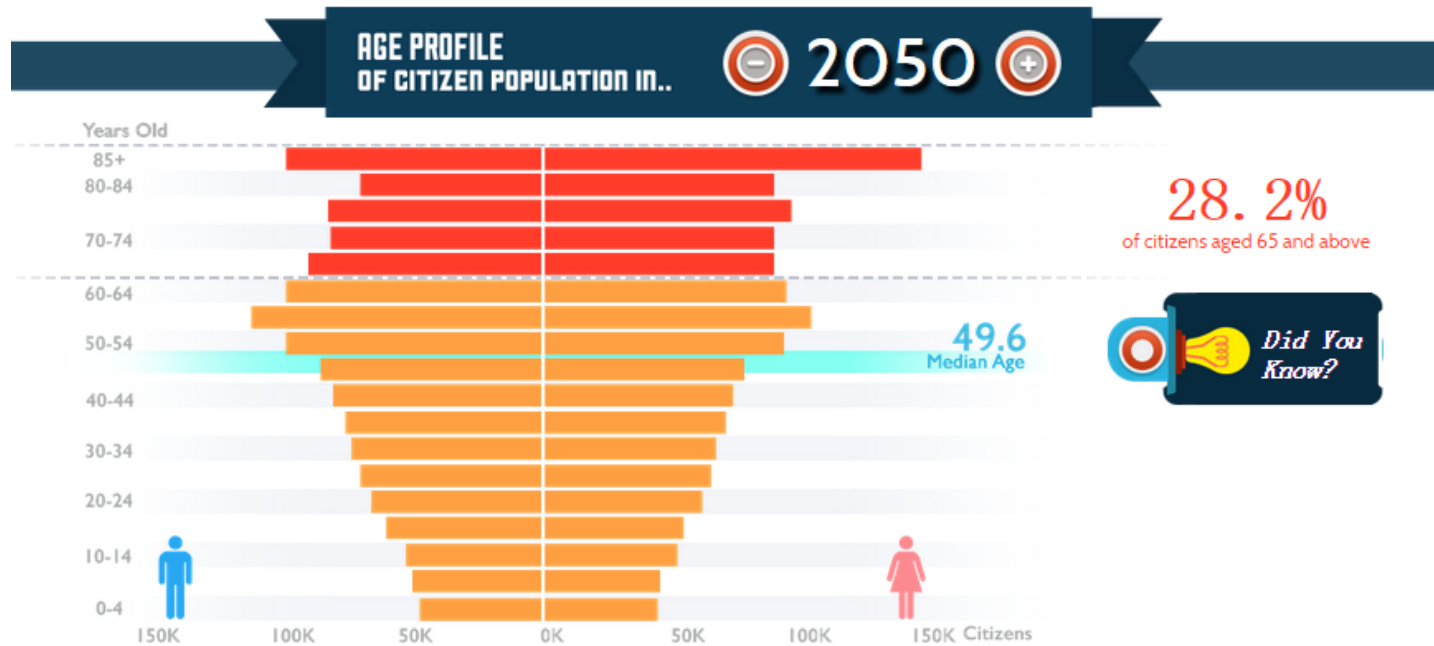
Assisting Patients

Stroke Rehabilitation
Daily living assistance

*We develop novel robotic systems, devices, and enabling technologies to address the healthcare and manpower challenges in societies with increasing **aging population**.*



Driving force: Aging Population



Source: Singapore Statistics



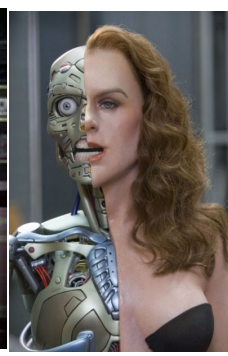
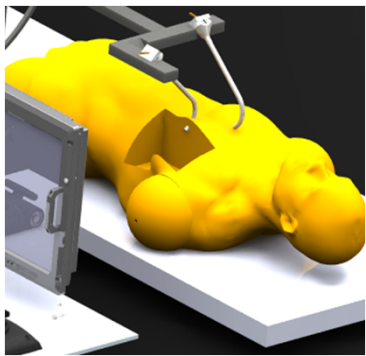
Resurgence of Robotics

- Demographic change
 - Ageing society
 - Escalating healthcare cost
 - Shrinking of young labor force
 - GDP growth from population increase to productivity gain
- Emerging new industry
 - AI and Robotics as rapidly growing industry



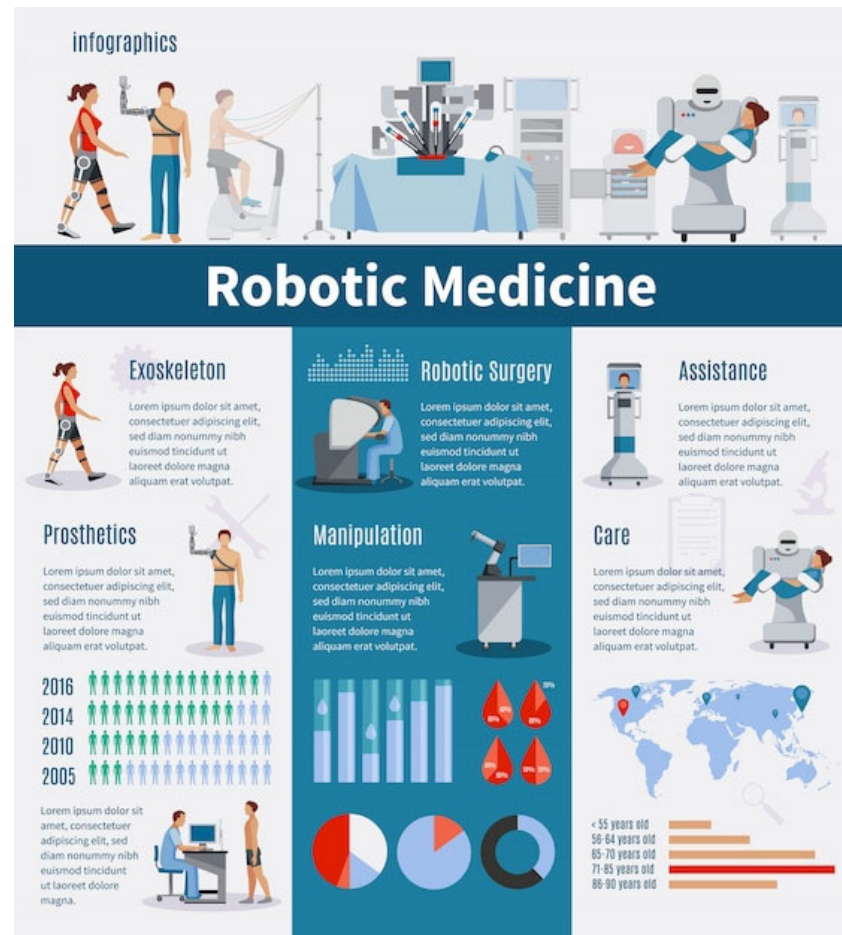
Robotics in Healthcare

- Surgical robots
- Prosthetic robots
- Rehabilitation robots
- Assistive Robots
- Social Robots



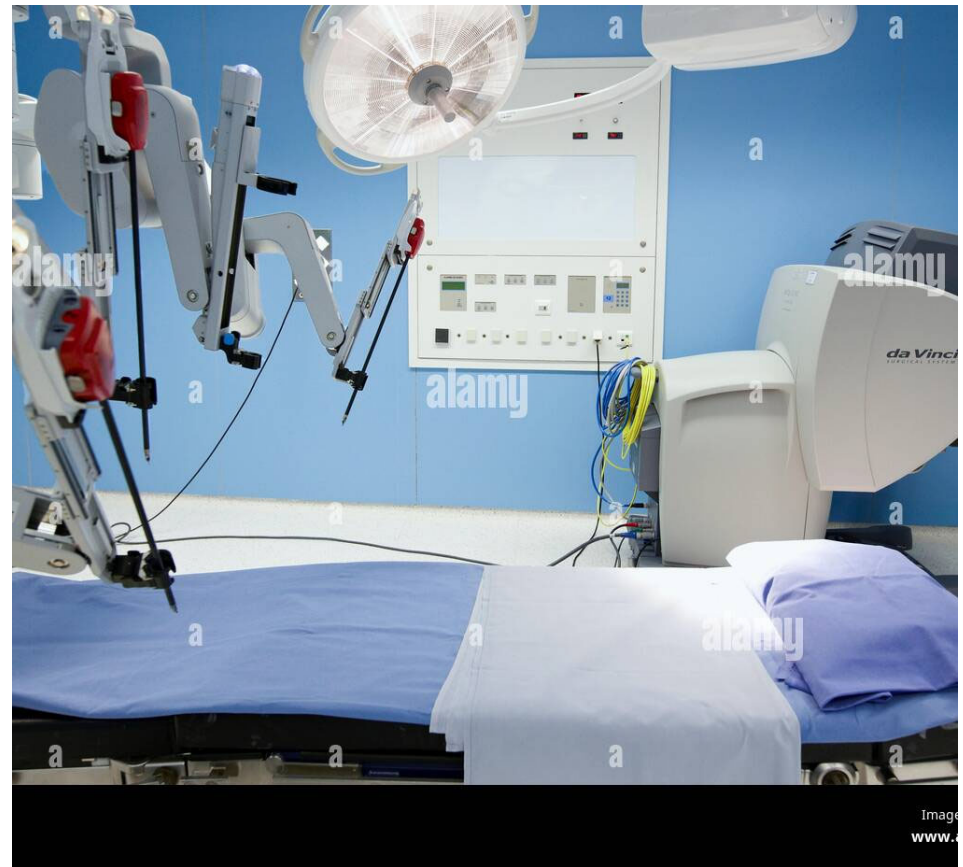
Applications of Medical Robotics?

- Medical robots assist in clinical tasks such as diagnosis, surgery, and rehabilitation.
- Core areas include:
 - Surgical robotics: Enhancing precision and safety.
 - Rehabilitation robotics: Supporting recovery from injuries or neurological conditions.
 - Assistive robotics: Providing mobility and independence.
 - Diagnostic robotics: Enabling faster and more accurate diagnoses.
 - Social robotics: Offering companionship and mental health support.



Evolution of Medical Robotics

- Key Milestones
 - 1990s:
 - FDA approval of the da Vinci Surgical System in 2000.
 - Emergence of rehabilitation robotics like MIT-Manus.
 - Expansion into new fields such as pediatric surgery and oncology.



Evolution of Medical Robotics

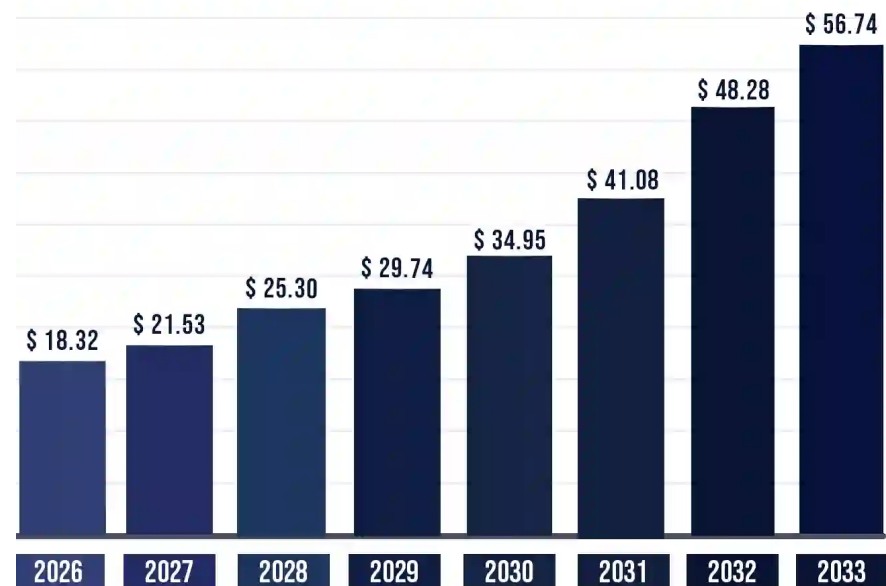
- Modern Developments
 - Integration of AI in diagnostics and navigation systems.
 - Use of cloud computing for data analysis and robotic learning.
 - Rise of wearable and portable robotic systems for home care.



Market Growth in Medical Robotics

- Global market projected to reach \$50 billion by 2030.
- Key growth factors:
 - Aging population requiring advanced care solutions.
- Increasing demand for minimally invasive surgeries

MEDICAL ROBOTS MARKET SIZE 2026 TO 2033 (USD BILLION)



www.precedenceresearch.com/medical-robots-market



THERE ARE 13 MAJOR
BRANDS OF ROBOTIC
SURGERY UNITS ON THE
MARKET TODAY

Surgical robots





Da Vinci Surgical System

- The **Intuitive Surgical** company developed the **da Vinci robotic system** to perform minimally invasive surgeries through superior visualization, enhanced dexterity, greater precision, and ergonomic comfort. With incisions of only 1 or 2 centimeters, surgeons can perform even complex procedures such as open-heart surgery. The system reduces hospital stays by half, reducing costs by about one-third, because of less pain and speedier recovery, according to the company.



Da Vinci Surgical System

- <https://www.youtube.com/watch?v=H4WzUSq9OeY>



Rehabilitation Engineering

The systematic application of engineering sciences to **design, develop, adapt, test, evaluate, apply, and distribute technological solutions** to problems confronted by individuals with **disabilities in functional areas**, such as **mobility, communications, hearing, vision, and cognition**, and in **activities associated with employment, independent living, education, and integration into the community**.

(USA's Rehabilitation Act of 1973)

Conventional Definition of rehabilitation engineering (RE)



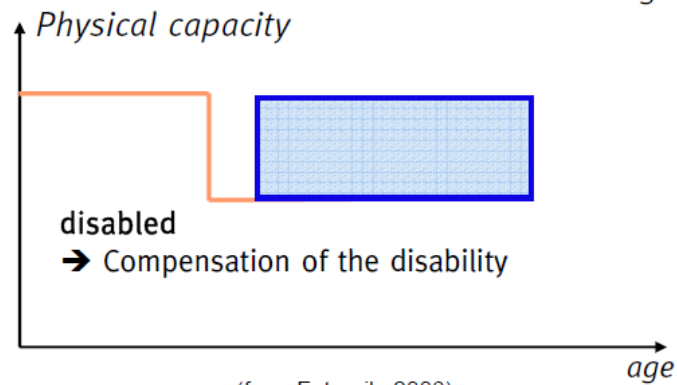
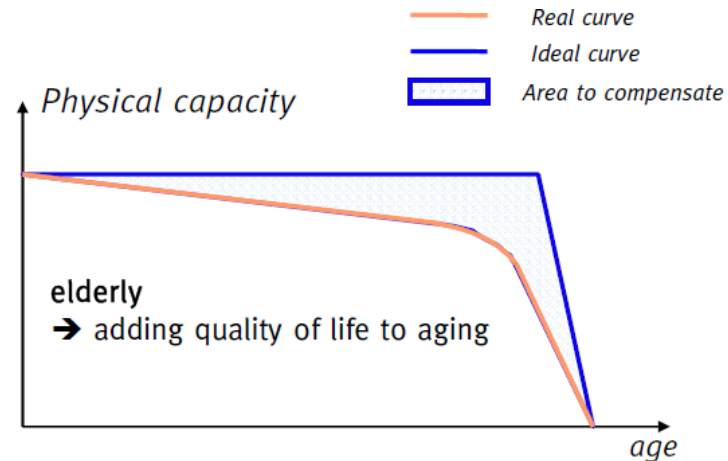
Assistive and Rehabilitation Devices

- Elderly
- Disabled

Physical capacity along time

- 1: detect the failing as soon as possible
- 2: compensate close to the 100% of capacity
- 3: continue the maximum of capacity as long as possible

Goal of AT: provide tools to improve or **restore** physical performance



(from Fatronik, 2006)



What is Rehabilitation

Rehabilitation is the process of helping an individual achieve the **highest level of function, independence, and quality of life** possible, with the aim of **integrating the individual into society**

Rehabilitation does **not reverse or undo** the **damage** caused by **disease or trauma**, but rather helps **restore the individual to optimal health, functioning, and well-being**



**Robotic Rehabilitation Center
Riley Hospital for Children, Indianapolis**



Rehabilitation engineers



iBOT

Roles

- Generally work in a team setting
- Collaboration with therapists, orthopedic surgeons, physical medicine specialists, neurologists

Areas of involvement

- ADLs, e.g. eating, toileting, combing hair, brushing teeth, reading
- Solutions to mobility
 - *Personal mobility: within home/office*
 - *Public (automobile, public transport, accessibility issues in buildings)*
 - *Ability to move functionally through environment*
 - *Reaching for object in a particular setting*
 - *Barriers that limit mobility: ill-fitted wheelchair cushion or support system*
- Solutions to sensory disabilities
 - *Sight, hearing, communications disorders (in both production side and comprehension side)*



Physical Disability

Main symptoms of Multiple sclerosis

Central:

- Fatigue
- Cognitive impairment
- Depression
- Unstable mood

Visual:

- Nystagmus
- Optic neuritis
- Diplopia

Speech:

- Dysarthria

Throat:

- Dysphagia

Musculoskeletal:

- Weakness
- Spasms
- Ataxia

Sensation:

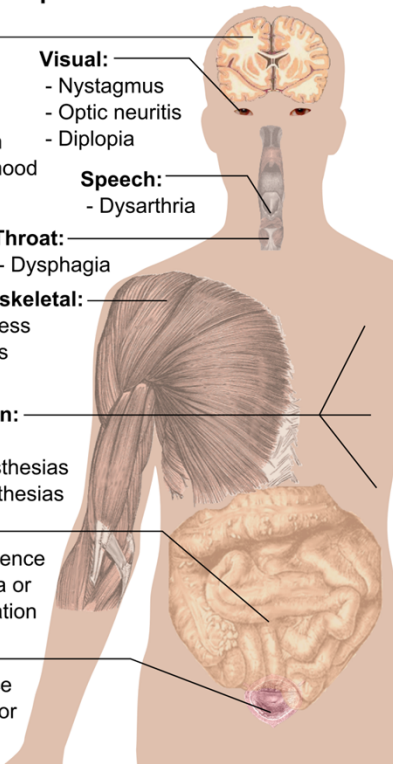
- Pain
- Hypoesthesias
- Paraesthesias

Bowel:

- Incontinence
- Diarrhea or constipation

Urinary:

- Incontinence
- Frequency or retention



Medical conditions

- Stroke, diabetes, muscular dystrophy, arthritis, cerebral palsy, multiple sclerosis..

Accidents

- Loss of limb
- Spinal cord injury
- Brain injury



Motor neuron disease



Common Areas of Assistive Technologies

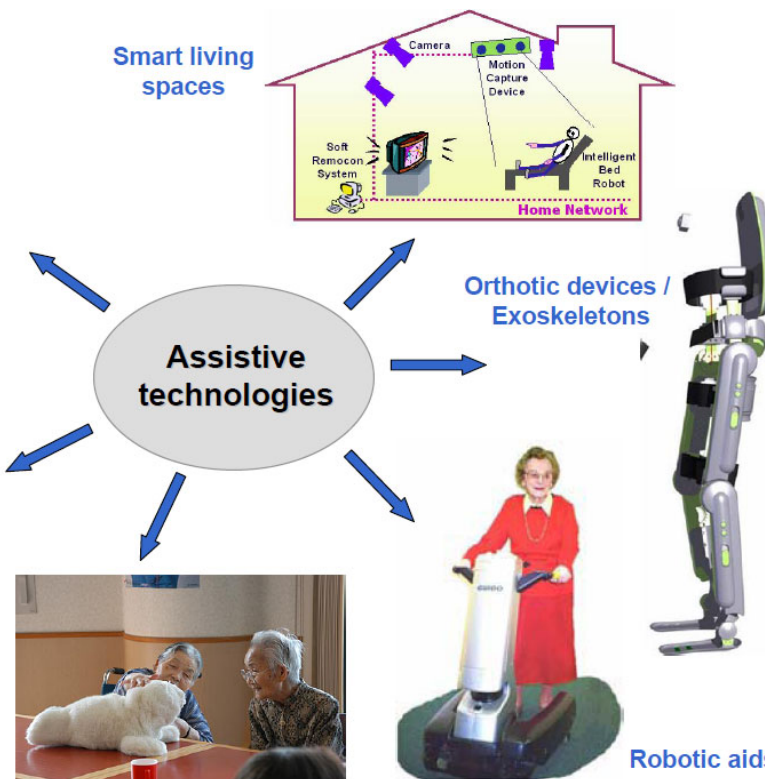


Prosthetic devices /
Artificial limbs

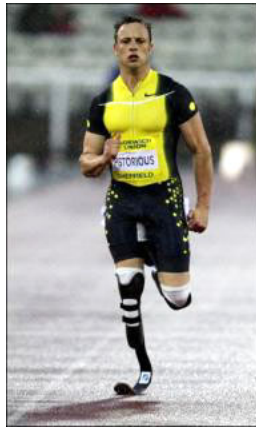


FES

Personal assistants



Loss of Limb



Artificial limb design

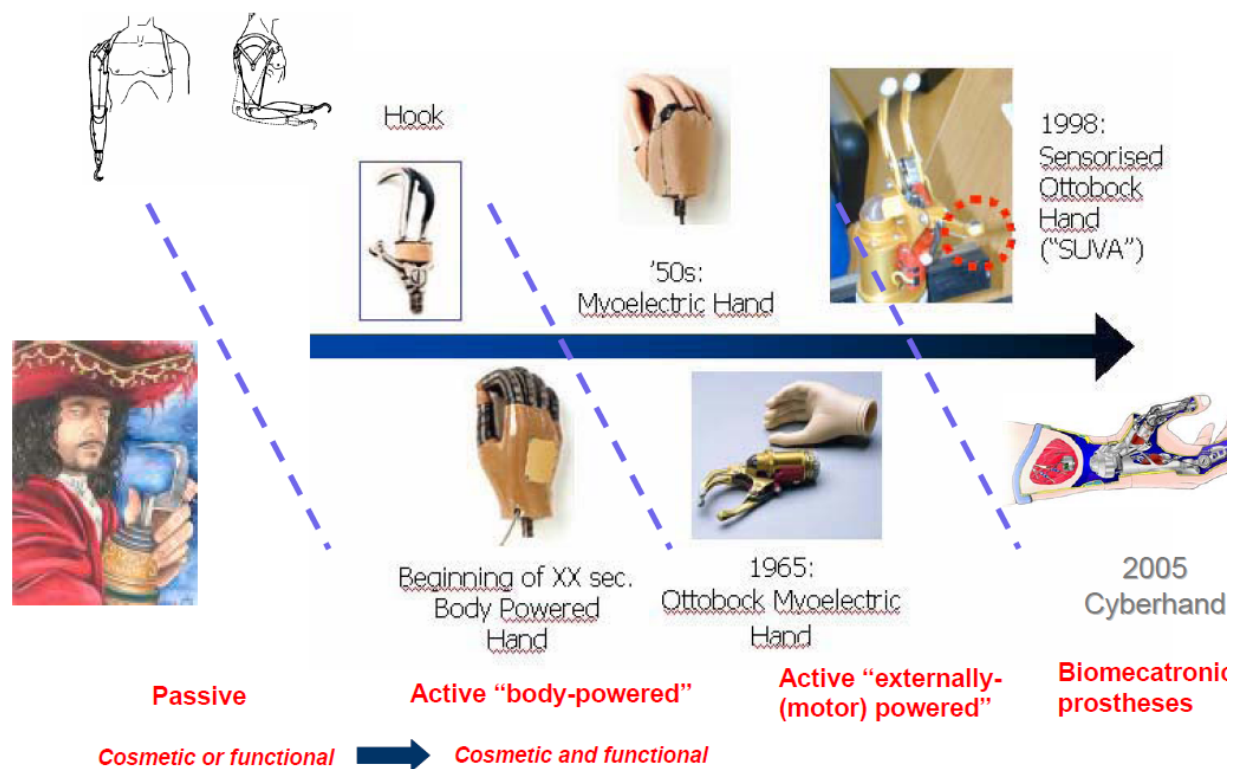
- Engineering complexity increases as amount of residual limb decreases, especially if one or more joints are lost
- Person with transtibial amputation could use simple wooden stump to extend leg, and could ambulate reasonably well

Advances in Prosthetics

- After WWII, concerted effort to foster better engineering design for artificial limbs in US
- **Dynamically lockable knee joints** were designed for above-knee amputees
- **Energy-storing artificial foot, Intelligent prosthesis**



Upper Limb Prosthetics



Evolution of the active hand prostheses (revisited from EURON Roadmap, 2004)

Upper Limb Prosthetics

Passive/cosmetic

- Look extremely natural
- Difficult to maintain, expensive
- Sacrifice function for cosmetic appearance

Functional

- Body-powered prostheses (*Cable-controlled*)
- Switch-controlled prostheses
- Myoelectric prostheses

Current research efforts

- Control of simultaneous multi-joint movement and provision for sensory feedback
- From end-effector of artificial arm to stump skin via electrical stimulation



Jesse Sullivan



Powered Upper Limb Prosthetics

- Types (Myoelectric (smart) upper-limb prosthetics)



(Myo prosthetics from Johns Hopkins)



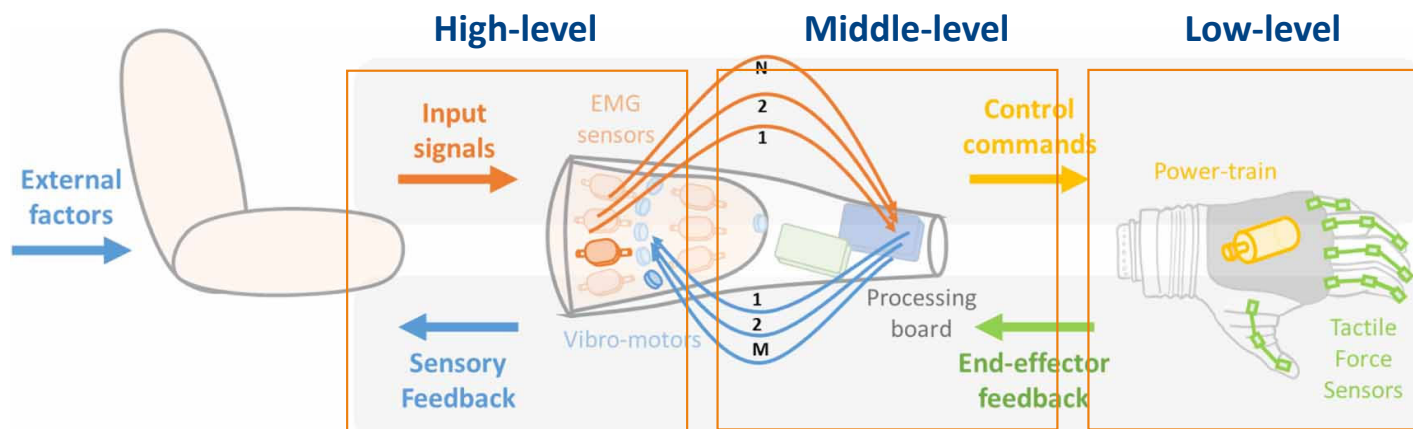
(Achilles prosthetics)

- Advantages: improve functionality, adaptability, potential for continuous improvement
- Disadvantages: *cost, weight, complexity, usage time*



Control of smart upper-limb prosthetics

- Hierarchical architecture of biological signal control



[Progress in Biomedical Engineering, 2022]

- **Hierarchical Architecture of biological signal control:** **lower-level** control: from control commands to motor actuation; **middle-level** control: from movement intention to control commands; **high-level** control: from input signals to movement intentions.



Challenges for powered prosthetics

- **User interface and control**

- The human arm/hand is a complicated, high DOF system
- Emphasis is on deducing user intent from motor pathway (muscles, nerves, brain) for natural control
- Fundamental control problem for prosthetics:
 - The higher the limb is amputated, more DOF need to be controlled by a prosthetic
 - But more of the limb nerve and muscle is missing, offering less control input



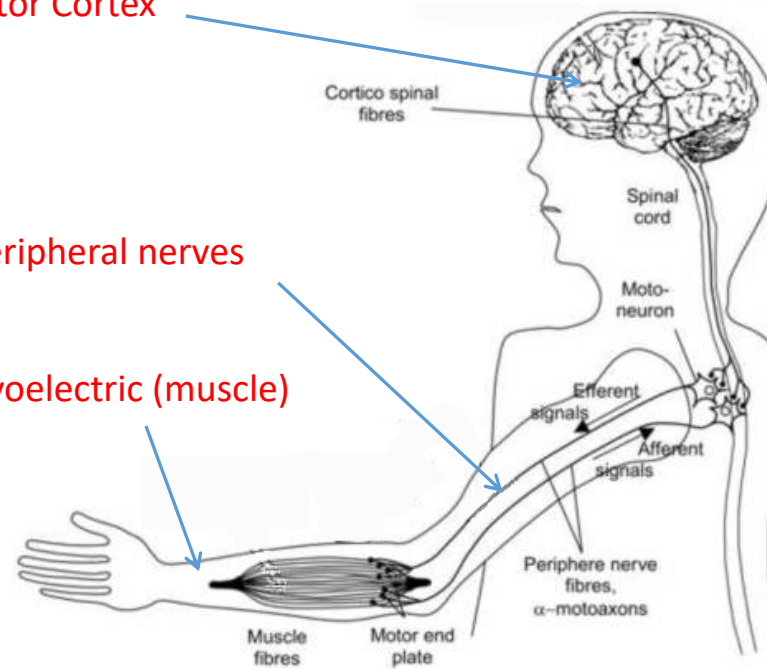
Challenges for powered prosthetics

Motor pathway (muscles, nerves, brain)

Motor Cortex

Peripheral nerves

Myoelectric (muscle)



EMG Based Control of Prosthetic Devices

- Appeal of EMG
 - Electrodes may be placed on surface of skin over muscle
 - Minimal risk, noninvasive, removable
 - Well-understood concept
- How can a multi-DOF prosthetic be controlled with limited remaining musculature?
 - Sharing control between user input and automation
 - (Manus hand)
 - Moving motor neurons to different muscles
 - (Kuiken 2006)



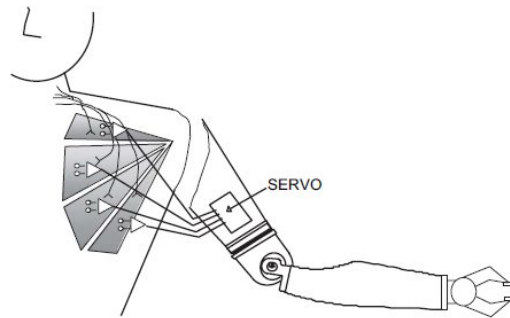
- **Myoelectrical-interface testing:** This testing detects the electrical signals from the residual muscles and evaluates the EMG activation mode for control of the smart upper-limb prosthetics.



Upper-limb Prosthetic Control

Targeted Muscle Reinnervation

Goal is to create more natural control of prosthetic limb while still using EMG as input by “re-wiring” motor neurons to intact muscle



1. Denervate an “expendable” area of muscle, such as in the chest.
2. Attach motor neurons from the amputated arm to different areas of the denervated chest muscles.
3. Use surface electrodes over chest muscles to provide many channels of EMG control input to prosthetic.

If reinnervation is successful, thinking about using the arm would result in contraction of the reinnervated muscles.

Upper-limb Prosthetic Control

Peripheral Nerve Control

Use of peripheral nerves as control input rather than EMG

Like TMR, an attempt to make limb control more intuitive than “coded” EMG

Sensory feedback can be sent through peripheral nerves, giving the user more control over joint position and grip force

Pulse train of varying current amplitude sent to electrodes in sensory nerves to re-create signals conveying touch and proprioception

Motor neuron firing rate decoded as force or torque command



Achievements:

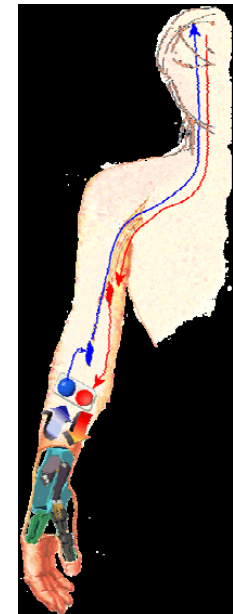
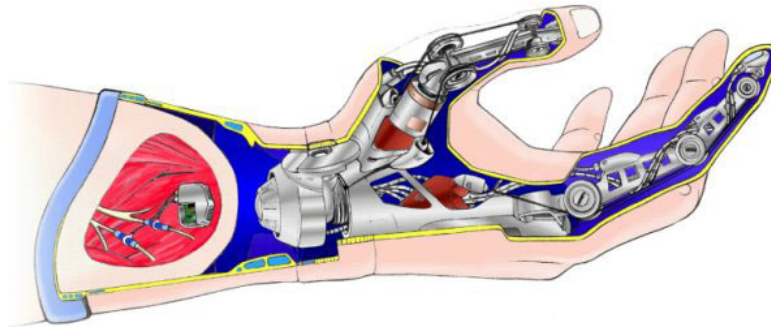
- Sensory information has potential to put human back in the loop during grasping
- Interpreting firing rate as torque or force command rather than trying to classify as a set motion

Shortcomings:

- Procedure involves implant to interface nerves to controller
- Pilot study was for single-DOF scenarios (grip force and elbow angle)

Upper-limb Prosthetic Control

Cybernetic prosthesis
(from the EU-FET “CYBERHAND” Project):
a cybernetic prosthesis controlled by the brain



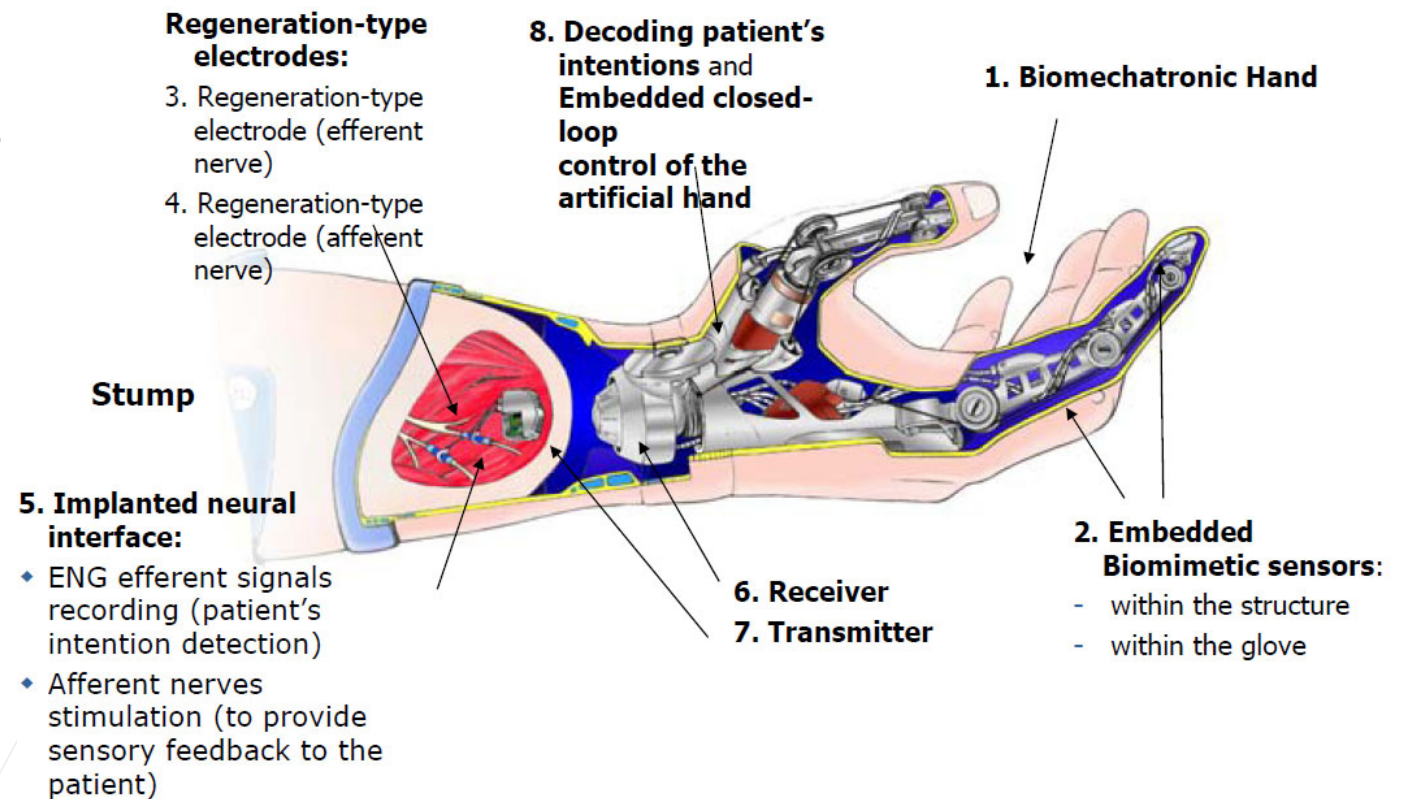
Neurobotics /prostheses : interfaces between an assistive device and the human nervous system such that the user's brain functions become part of the system control loop

R&D issues

- biocompatible implantable micro-sensors and electrodes
- Neurophysiology: which neurons to interface?
- Data processing (volitional command and artificial sensory feedback)...

Upper-limb Prosthetic Control

- **CyberHand Architecture**





Cybernetic prosthesis



Scuola Superiore
Sant'Anna

di Studi Universitari e di Perfezionamento

ARTS Lab

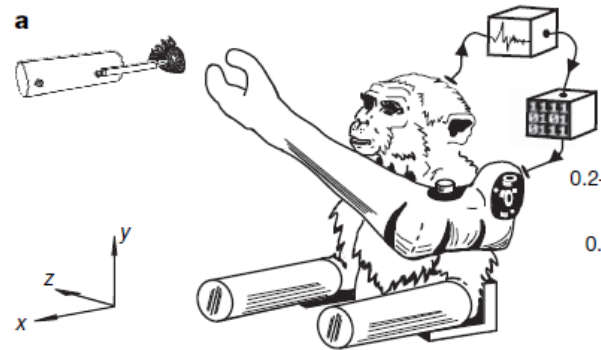
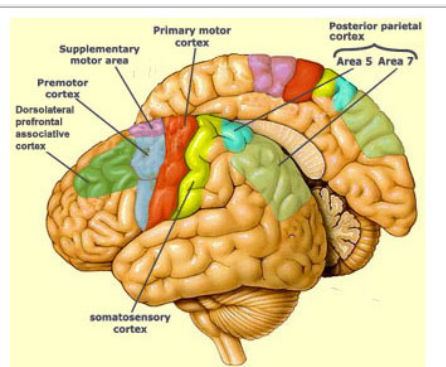
Advanced Robotics Technology and Systems



Upper-limb Prosthetic Control

Cortical Control

- Upper limb motion representation in the brain:
 - Certain region of primary motor cortex associated with arm movement
 - Speed and direction of desired motion can be inferred from neural activity
 - Population Vector Hypothesis (Georgopoulos 1988)
 - Each neuron has a preferred firing direction; activity of that neuron indicates motion in preferred direction.
 - Represent velocity of hand as vector sum of preferred firing directions weighted by firing rate



Upper-limb Prosthetic Control

Cortical Control



<https://www.youtube.com/watch?v=ogBX18maUiM>



Advanced Technologies for Prosthetic Devices

Neuroengineering

Neuroengineering is defined as the interdisciplinary field of engineering and computational approaches to problems in basic and clinical neurosciences.

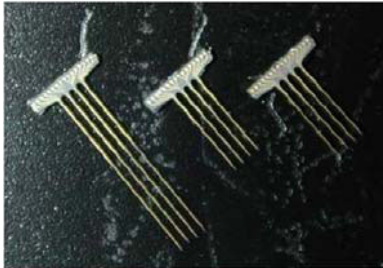
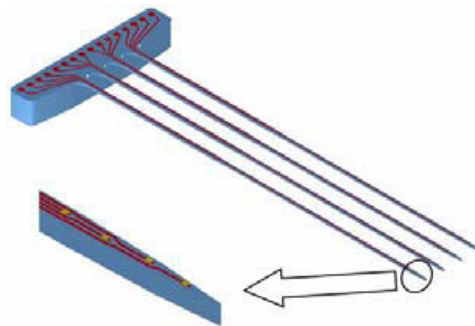
Neuroengineering encompasses the fields of

- Engineering
- Mathematics
- Computer science
- Molecular and cellular systems
- Neurosciences



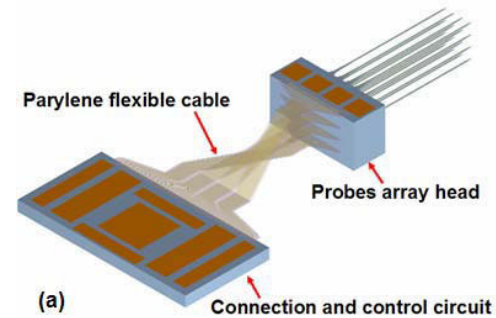
Advanced Technologies for Prosthetic Devices

Steps to Neural Prosthesis: Implanted Microelectrode Arrays



Under Development at U of
Michigan, U of Utah and Cal
Tech

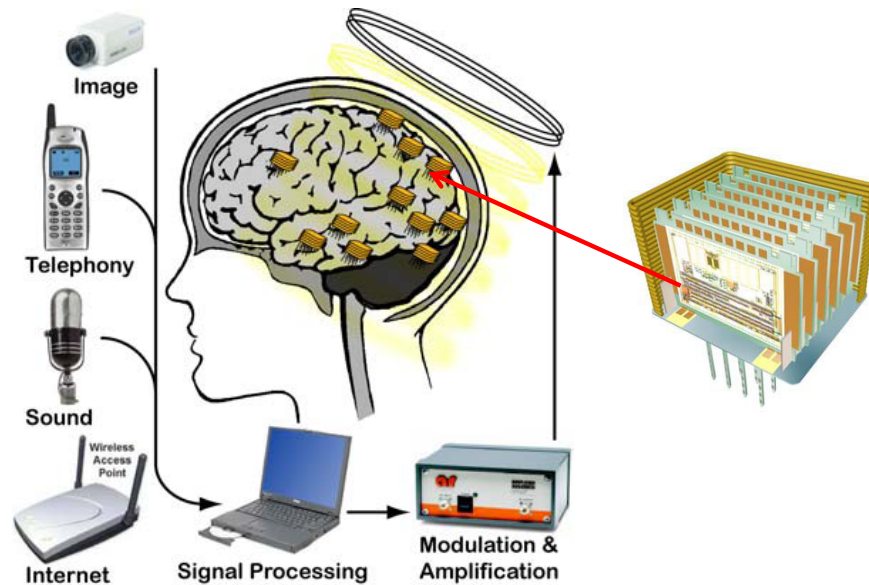
Each shank on each probe
has four contacts.



Multiple probes form a
3-D geometry

Advanced Technologies for Prosthetic Devices

Steps to Neural Prosthesis: A Distributed Wireless 3-D Implants



Neuralink



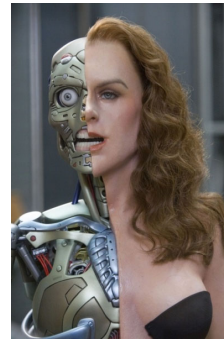
- https://www.youtube.com/watch?v=Gv_XB6Hf6gM

Advanced Technologies for Prosthetic Devices

Current



Future



Future Applications



Future of Assistive and Rehabilitation Robotics Technologies



Disability is still prevalent, particularly with ageing population

Engineering disciplines permeate many aspects of rehabilitation:

ECE (Signal processing, control/information theory, neural networks, micro-fabrication, fuzzy logic, virtual reality, image processing...), **BME** (Biomechanical testing, gait/motion analysis, prosthetic fitting, seat cushion and back support design, and design of artificial joints), **Material E** (Newer biocompatible materials), **Chemical E** (Implantable sensors), **Industrial E** (rehabilitative ergonomics)

RE challenge is to find advances in any field, engineering or otherwise, that aid disabled clients to function independently with skills to integrate into society



Course Focused Topics (Part 2 by Prof. Yu)

- Stroke and Stroke Rehabilitation
- Robotics for upper limb rehabilitation
- Robotics for upper lower rehabilitation





Summary

- Driving forces of Robotics in Healthcare
- Overview of various rehab and assistive robotics
- Outline of course focused topics (Part 2)

